

Watching the Money, Missing the Message: Covert Communication Risks in Blockchain Transactions

Introduction

Recent investigations into terrorist financing indicate a sustained and growing reliance on virtual assets. TRM Labs reported a 240 percent increase in Tether (USDT) usage among tracked terror-financing entities in 2022. A subsequent investigation identified pro-Islamic State Khorasan Province (ISKP) networks in Tajikistan channeling approximately US\$2 million toward recruitment activities. In March 2025, the United States Department of Justice disrupted a Hamas-linked cryptocurrency network that had accumulated over US\$1.5 million. The Financial Action Task Force (FATF) further noted that stablecoins accounted for 84 percent of illicit virtual asset volume in its 2026 assessment.

These developments demonstrate that while financial surveillance mechanisms have improved, they remain focused on tracing monetary flows. They do not account for the possibility that blockchain transactions may also function as covert communication channels. This analytical gap motivates the present study.

The Problem: Beyond Financial Surveillance

Blockchain monitoring systems are designed to identify and trace financial activity. However, they are not equipped to detect non-financial information embedded within transactions. This limitation is structural rather than operational. Even highly advanced blockchain analytics platforms focus on transaction flows, wallet clustering, and anomaly detection within financial parameters.

The emergence of on-chain steganography challenges this model. This technique allows actors to embed encrypted information within transaction data in a manner that remains indistinguishable from legitimate activity. For example, a minimal-value transfer—such as

US\$0.01 in USDT—can carry encoded instructions within cryptographic components of the transaction without altering its observable financial characteristics.

This creates a parallel communication layer that operates within monitored infrastructure but remains analytically invisible.

The Technical Gap

A comprehensive survey in IEEE Communications Surveys and Tutorials identifies four primary covert communication channels within blockchain systems: address-based, value-based, script-based, and signature-based channels. Of these, three exhibit statistical patterns that can be detected under sufficient analytical scrutiny.

The fourth, known as the subliminal signature channel, presents a fundamentally different challenge. It exploits the nonce used in Elliptic Curve Digital Signature Algorithm (ECDSA)-based transactions. This nonce must appear random for the signature to remain valid. By deriving the nonce deterministically from encoded message bits and a shared secret, actors can embed information without altering the statistical properties of the signature.

Research presented at IEEE EuroS&P demonstrated the practical implementation of this technique on the Bitcoin test network. The findings confirmed that such communication remains indistinguishable from legitimate cryptographic randomness under standard assumptions. With an effective bandwidth of approximately 256 bits per transaction, this method is sufficient to transmit operational signals such as encryption keys, geolocation coordinates, or execution commands.

Why Existing Systems Miss the Threat

Current monitoring frameworks are not designed to analyze cryptographic entropy at the transaction level. Instead, they rely on observable financial indicators such as transaction volume, frequency, and network relationships. A single low-value transaction carrying embedded data does not generate anomalies in any of these dimensions.

Empirical evaluations reinforce this limitation. A 2025 USENIX Security study of commercial blockchain analytics platforms found that non-financial data embedded within transactions falls outside their analytical scope. This is not a technological deficiency but a consequence of compliance-driven design priorities.

At the same time, blockchain usage patterns indicate a shift among illicit actors. Following enforcement actions, Hamas suspended public Bitcoin fundraising in April 2023 after Israeli asset seizures. This shift has often been interpreted as a withdrawal from cryptocurrency use. However, a more plausible interpretation is strategic adaptation: separating

financial transactions from communication channels to reduce exposure.

Evidence of Operational Feasibility

The feasibility of blockchain-based covert communication has already been demonstrated outside terrorist contexts. Akamai’s Security Intelligence Response Team documented a case in which the Skidmap botnet encoded command-and-control information within Bitcoin transactions. Although this implementation used a more detectable channel, it was not identified through blockchain monitoring systems but through endpoint detection methods.

The subliminal signature channel presents a more sophisticated alternative with no equivalent detection surface. As detection capabilities evolve, adversaries are likely to adopt increasingly covert techniques.

The relevance of this risk is amplified by network preferences. The United Nations Office on Drugs and Crime (UNODC) identified Tron as a primary platform for cryptocurrency-based money laundering in Asia in 2024. TRM Labs’ 2025 report estimated that Tron accounted for 58 percent of illicit crypto activity. Despite this, no systematic steganalysis of Tron transaction data has been conducted.

India’s Structural Blind Spot

India has demonstrated growing capacity in tracking cryptocurrency-based terror financing. In 2023, the National Investigation Agency (NIA) filed charges against an Islamic State-linked operative in Bihar who routed funds through cryptocurrency to handlers in Syria. A separate case revealed in Karnataka High Court proceedings in 2026 involved cryptocurrency transfers used to support reconnaissance and arson activities in Mangaluru.

These cases highlight effective financial surveillance. However, they also reveal a limitation: detection depends on observable monetary flows. Communications embedded within transactions would not trigger investigative thresholds.

Institutionally, relevant capabilities exist but remain fragmented. The Financial Intelligence Unit–India (FIU-IND) maintains transaction data. The National Technical Research Organisation (NTRO) possesses cryptographic expertise. The NIA provides investigative context. However, no coordinated framework currently treats transaction-level cryptographic data as an intelligence source.

Conceptual Clarification: Type I and Type II Dual-Use

To address this gap, it is necessary to distinguish between two categories of dual-use blockchain data.

Type I dual-use refers to incidental or residual data generated as part of normal protocol operations. Existing monitoring systems partially capture this category.

Type II dual-use refers to intentional embedding of covert communications within transaction data. This category is not addressed by current frameworks and requires a distinct analytical approach.

The distinction is not merely conceptual. It determines whether monitoring systems are focused on the appropriate layer of analysis. Type I detection relies on pattern recognition and anomaly detection. Type II detection requires cryptographic entropy analysis and transaction-level examination.

Policy Recommendations

1. Establish International Norms

India should take a leadership role within the Asia/Pacific Group on Money Laundering (APG) to incorporate Type II classifications into FATF guidance. This would establish a global framework distinguishing between incidental and intentional data embedding.

2. Develop Domestic Analytical Capability

A joint mandate should be established between FIU-IND, NTRO, and the NIA to conduct steganalysis of blockchain transactions. Analytical triggers may include deviations in nonce entropy relative to network baselines, combined with proximity to known high-risk entities.

Legal clarity will be required to determine whether existing provisions under the Prevention of Money Laundering Act (PMLA) extend to cryptographic signature data. If not, data-sharing mechanisms must be formalized.

3. Build Empirical Datasets

India should prioritize the development of a baseline entropy dataset for high-risk networks such as Tron. Institutions including IIT Kanpur, CERT-In, and C-DAC possess the technical expertise required for large-scale analysis but currently lack a unified mandate.

Generating empirical evidence would position India at the forefront of global research in this domain.

Conclusion

The asymmetry between risk and response is significant. Developing detection capabilities for Type II covert channels requires limited resources relative to the potential consequences of inaction.

The techniques discussed are not speculative. They have been documented in open academic literature and demonstrated in operational contexts. As financial surveillance improves, adversaries are incentivized to shift toward communication methods that evade detection.

The absence of evidence should not be interpreted as evidence of absence. A system that monitors transactions without examining embedded data risks overlooking a critical dimension of threat activity.

References

References

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